Xiaoyin Chen

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Education

Mila, Université de Montreal — Sept. 2023 – Present
Doctor of Philosophy, Computer Science, Advisor: Prof. Yoshua Bengio — *GPA: 4.00/4.00*

Duke University — Aug. 2021 – May 2023
Master of Science, Computer Science, Advisor: Prof. Sam Wiseman — *GPA: 4.00/4.00*

University of California, Irvine — Sept. 2016 – Mar. 2020
Bachelor of Science, Computer Science, Minor in Statistics — *GPA: 3.94/4.00, Major GPA:3.99/4.00*

Work Experience

Student Researcher — Jun. 2026 – Present
Google DeepMind, Mountain View, USA

Research Intern — Sep. 2025 – Jun. 2026
Snowflake Research, Menlo Park, USA

Visiting Researcher — Nov. 2024 – Jul. 2025
ServiceNow Research, Montreal, Canada

Publications & Preprints

Peer-reviewed conference, journal, and workshop

Learning to Self-Evolve
Xiaoyin Chen, Canwen Xu, Yite Wang, Boyi Liu, Zhewei Yao, Yoshua Bengio, Yuxiong He.
ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving, https://arxiv.org/abs/2603.18620, 2026.

How Much Backtracking is Enough? Exploring the Interplay of SFT and RL in Enhancing LLM Reasoning
Hongyi James Cai, Junlin Wang, **Xiaoyin Chen**, Bhuwan Dhingra.
Conference on Language Modeling (COLM) 2026, openreview.net/forum?id=bpVWLDwlu4, 2026.

Self-Evolving Curriculum for LLM Reasoning
Xiaoyin Chen, Jiarui Lu, Minsu Kim, Dinghuai Zhang, Jian Tang, Alexandre Piché, Nicolas Gontier, Yoshua Bengio, Ehsan Kamalloo.
Conference on Language Modeling (COLM) 2026, https://arxiv.org/abs/2505.14970, 2025.

Bayesian Symbolic Regression with Entropic Reinforcement Learning
Oussama Boussif, Mohammed Mahfoud, Younesse Kaddar, Moksh Jain, Sida Li, Damiano Fornasiere, **Xiaoyin Chen**, Yoshua Bengio, Esmeralda S. Whitammer.
Uncertainty in Artificial Intelligence (UAI) 2026, https://arxiv.org/abs/2608.09617, 2026.

When Greedy Wins: Emergent Exploitation Bias in Meta-Bandit LLM Training
Sanxing Chen, **Xiaoyin Chen**, Yukun Huang, Roy Xie, Bhuwan Dhingra.
International Conference on Learning Representations (ICLR) 2026, https://arxiv.org/abs/2509.24923, 2025.

Search-Based Correction of Reasoning Chains for Language Models
Minsu Kim, Jean-Pierre Falet, Oliver E. Richardson, **Xiaoyin Chen**, Moksh Jain, Sungjin Ahn, Sungsoo Ahn, Yoshua Bengio.
International Conference on Learning Representations (ICLR) 2026, https://arxiv.org/abs/2505.11824, 2025.

PipelineRL: Faster On-policy Reinforcement Learning for Long Sequence Generation
Alexandre Piché, Ehsan Kamalloo, Rafael Pardinas, **Xiaoyin Chen**, Dzmitry Bahdanau.
Transactions on Machine Learning Research (TMLR), https://arxiv.org/abs/2509.19128, 2025.

Aligning Protein Conformation Ensemble Generation with Physical Feedback
Jiarui Lu*, **Xiaoyin Chen***, Stephen Zhewen Lu, Aurelie Lozano, Aurelie Lozano, Vijil Chenthamarakshan, Payel Das, Jian Tang.
Forty-Second International Conference on Machine Learning, https://arxiv.org/abs/2505.24203, 2025.

Structure Language Models for Protein Conformation Generation
Jiarui Lu*, **Xiaoyin Chen***, Stephen Zhewen Lu, Chence Shi, Hongyu Guo, Yoshua Bengio, Jian Tang.
International Conference on Learning Representations 2025, https://arxiv.org/abs/2410.18403, 2024.

HarmAug: Effective Data Augmentation for Knowledge Distillation of Safety Guard Models
Seanie Lee, Haebin Seong, Dong Bok Lee, Minki Kang, **Xiaoyin Chen**, Dominik Wagner, Yoshua Bengio, Juho Lee, Sung Ju Hwang.
International Conference on Learning Representations 2025, https://arxiv.org/abs/2410.01524, 2024.

Proof Flow: Preliminary Study on Generative Flow Network Language Model Tuning for Formal Reasoning
Matthew Ho, Vincent Zhu, **Xiaoyin Chen**, Moksh Jain, Nikolay Malkin, Edwin Zhang.
System-2 Reasoning at Scale Workshop at NeurIPS 2024, https://arxiv.org/abs/2410.13224, 2024.

Inference and Verbalization Functions During In-Context Learning
Junyi Tao*, **Xiaoyin Chen***, Nelson F. Liu.
Findings of Empirical Methods in Natural Language Processing (EMNLP) 2024, https://arxiv.org/abs/2410.09349, 2024.

Efficient Causal Graph Discovery Using Large Language Models
Thomas Jiralerspong*, **Xiaoyin Chen***, Yash More, Vedant Shah, Yoshua Bengio.
AGI Workshop at ICLR 2024, https://arxiv.org/abs/2402.01207, 2024.

Learning Consistent Deep Generative Models from Sparse Data via Prediction Constraints
Gabriel Hope, Madina Abdrakhmanova, **Xiaoyin Chen**, Michael C. Hughes, Erik B. Sudderth.
4th Symposium on Advances in Approximate Bayesian Inference, 2022. arxiv.org/abs/2012.06718.

On the Current Failure – But Bright Future – of Topology-driven Biological Network Alignment
Siyue Wang, **Xiaoyin Chen**, Brent J. Frederisy, Benedict A. Mbakogu, Amy D. Kanne, Pasha Khosravi, Wayne B. Hayes.
Advances in Protein Chemistry and Structural Biology: Protein interaction networks, Volume 131.
https://arxiv.org/abs/2204.11999.

Cross-species Prediction of Protein Function by Global Network Alignment
Siyue Wang, **Xiaoyin Chen**, Brent J. Frederisy, Benedict A. Mbakogu, Amy D. Kanne, Pasha Khosravi, Giles R.S. Atkinson, Wayne B. Hayes.
28th Conference on Intelligent Systems for Molecular Biology (protein prediction track), 2020.
https://www.iscb.org/cms_addon/conferences/ismb2020/tracks/functioncosi.

Preprints & Under Review

BM25 Query Augmentation Learned End-to-End
Xiaoyin Chen, Sam Wiseman.
https://arxiv.org/abs/2305.14087, 2023.

Teaching Experience

Teaching Assistant
Duke University
- **CS590.05 Spring 2023**: Computational Biology.

Undergrad Tutor & Grader
University of California, Irvine
- **ICS-33 Spring 2017**: Intermediate Programming in Python;
- **ICS-46 Fall 2018**: Data Structure Implementation and Analysis in C++.

Service

Reviewer: ACL, EMNLP, Neurips, ICLR, ICML